



Southern Nuclear

E. D. Dean III
Vice President – Plant Hatch

Plant Edwin I. Hatch
11028 Hatch Parkway North
Baxley, GA 31513
912.537.5859 tel
912.366.2077 fax

OCT 16 2019

Docket Nos.: 50-366

NL-19-1264

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant Unit 2
Licensee Event Report 2019-004-00
High Pressure Coolant Injection System Inoperable Due to Brief Loss of Power to HPCI Pump
Flow Controller

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(v)(D), Southern Nuclear Operating Company hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact the Hatch Licensing Manager, Jimmy Collins at 912.537.2342.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'E. D. Dean III' with a stylized flourish at the end.

E. D. Dean
Vice President – Hatch

EDD/JEL/SCM

Enclosure: LER 2019-004-00

Cc: Regional Administrator, Region II
NRR Project Manager – Hatch
Senior Resident Inspector – Hatch
RTYPE: CHA02.004

Edwin I. Hatch Nuclear Plant Unit 1 and 2

**Licensee Event Report 2019-004-00
High Pressure Coolant Injection System Inoperable Due to Brief Loss of Power to HPCI
Pump Flow Controller**

Enclosure

LER 2019-004-00



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Edwin I. Hatch Nuclear Plant Unit 2

2. Docket Number

05000

366

3. Page

1 OF

3

4. Title

High Pressure Coolant Injection System Declared Inoperable Due to Brief Loss of Power to HPCI Pump Flow Controller

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
08	22	2019	2019	004	00	10	16	2019	Facility Name	Docket Number
9. Operating Mode										
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)										
1			☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)				
			☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)				
			☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)				
			☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)				
10. Power Level			☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)				
100			☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)				
			☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)				
			☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)				
			☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)				
			☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)						

12. Licensee Contact for this LER

Licensee Contact

Jimmy Collins - Licensing Manager

Telephone Number (Include Area Code)

912-537-2342

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	BJ	JX	L045	Y					

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On August 22, 2019 at 2234 EST with Unit 2 operating at 100% rated thermal power, the "High Pressure Coolant Injection (HPCI) system inverter circuit failure" alarm was received on a Main Control Room panel. This alarm indicates a loss of AC power from the HPCI inverter, and is caused by either a loss of DC power to, or a failure of, the inverter. Operating personnel immediately noticed the loss of indication on several digital recorders and on the HPCI flow controller. At 2235 EST, the "HPCI system inverter circuit failure" alarm cleared, and all referenced digital indications returned to normal.

Subsequent investigation into the HPCI inverter circuit failure revealed a failed ceramic capacitor in the HPCI turbine speed control power supply. The failed capacitor caused a short-circuit condition that momentarily drew power down on the HPCI inverter. This caused a momentary loss of indication to the referenced digital recorders and a momentary loss of power to the HPCI flow controller. In the event of a valid HPCI initiation signal, the momentary loss of power to the HPCI flow controller would have resulted in a loss of HPCI function. Operations personnel therefore declared HPCI inoperable at 2234 EST.

The cause of the capacitor failure was not determined. Ceramic capacitors of this type are not known to fail at a specified frequency and are not recommended for replacement on a periodic basis. The HPCI turbine speed control power supply was subsequently removed from service, replaced, and the new power supply tested satisfactorily.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Edwin I. Hatch Nuclear Plant Unit 2	05000- 366	YEAR	SEQUENTIAL NUMBER	REV NO.
		2019	004	00

NARRATIVE**EVENT DESCRIPTION**

On August 22, 2019 at 2234 EST with Unit 2 operating at 100% rated thermal power, the "High Pressure Coolant Injection (HPCI) system inverter circuit failure" alarm was received on a Main Control Room panel. Operating personnel immediately noticed the loss of indication on several digital recorders and on the HPCI flow controller. At 2235 EST, the "HPCI system inverter circuit failure" alarm cleared and all digital indications returned to normal. At 2234 EST, HPCI was declared inoperable due to the failure of a ceramic capacitor inside of the HPCI turbine speed control power supply. The failed capacitor caused a short-circuit condition that momentarily drew down power on the HPCI inverter and resulted in a loss of function to the HPCI flow controller.

Failed Component Information:

Master Parts List Number: 2E41K065
Manufacturer: Lambda Electronics
Model Number: LRS 52-48
Type: Power Supply

The capacitor failure would have ultimately resulted in a loss of HPCI function during a demand and rendered HPCI inoperable.

EVENT CAUSE ANALYSIS

Upon investigation into the "HPCI system inverter circuit failure" alarm, a loss of indication was noticed on several digital indicators and a burning smell was noted on one of the Main Control Room back panels. Subsequent visual inspection revealed a failed ceramic capacitor in the HPCI turbine speed control power supply. The cause of the capacitor failure was not determined. Ceramic capacitors of this type are not known to fail at a specific frequency and are not recommended for replacement on a periodic basis.

SAFETY ASSESSMENT

This event is reportable per 10 CFR 50.73(a)(2)(v)(D) due to an event or condition that could have prevented the fulfillment of the safety function of a structure or system that is needed to mitigate the consequences of an accident.

The HPCI system is a single-train system that is designed to ensure that the reactor is adequately cooled to limit fuel-cladding temperature in the event of a small break in the nuclear system and a loss of coolant does not result in a rapid reactor pressure vessel depressurization. The HPCI system will continue to operate until the Reactor Pressure Vessel (RPV) is below the pressure at which either operation of the Low Pressure Coolant Injection (LPCI) mode of the Residual Heat Removal (RHR) system or the Core Spray (CS) system will maintain core cooling.

Upon a small break which results in a loss of coolant accident (LOCA), HPCI provides sufficient inventory to prevent the core from becoming uncovered. In the event that HPCI is unavailable or inoperable, the Automatic Depressurization System (ADS) is credited to depressurize the RPV to enable operation of the low pressure ECCS systems, i.e., CS or LPCI. Reactor Core Isolation Cooling (RCIC) is also available to restore and maintain water level.

Although HPCI was declared inoperable, ADS, CS, LPCI, and RCIC were operable and could have provided core cooling in the event of a small break LOCA or a Design Basis Accident (DBA) LOCA. Therefore, based on the ability of the station to mitigate the consequences of a potential loss of the HPCI system combined with a small break LOCA, this event is considered to have very low safety significance.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Edwin I. Hatch Nuclear Plant Unit 2	05000- 366	YEAR 2019	SEQUENTIAL NUMBER 004	REV NO. 00

NARRATIVE

CORRECTIVE ACTIONS

The cause of the capacitor failure was not determined. Ceramic capacitors of this type are not known to fail at a specific frequency, and are not recommended for replacement on a periodic basis.

The HPCI turbine speed control power supply was removed from service, replaced, and the new power supply tested satisfactorily.

PREVIOUS SIMILAR EVENTS

A similar event, in which the Unit 1 HPCI system was declared inoperable due to a degraded inverter, occurred on February 8, 2017. The HPCI inverter supplies power to the HPCI flow controller and power supply. HPCI was declared inoperable due to the degraded voltage condition. The degraded inverter was removed from service and replaced, and HPCI was returned to operable status. It should be noted that the only similarity between the event on February 8, 2017 and the referenced event is the degraded voltage condition. The causes of the degraded voltage condition were different.